

Polyamide 66

with carbon fibers, heat stabilized, natural color
(black)

Physical properties		Test method	Specimen	Units	Typical value
Specific gravity		ISO 1183-1		g/cm ³	1,20
Water absorption	23°C / 24h	ISO 62	MPTS ISO 3167 A	%	<1,0
Linear mould shrinkage		DIN 16742	MPTS ISO 3167 A	%	0,2-0,5
Mechanical properties at 23°C / 50% rh					
Tensile strength	dry, @50 mm/min	ISO 527	MPTS ISO 3167 A	MPa	240
Elongation at maximum force	dry, @50 mm/min	ISO 527	MPTS ISO 3167 A	%	2,1
Modulus of elasticity	dry, @1 mm/min	ISO 527	MPTS ISO 3167 A	GPa	15,5
Flexural strength	dry, @10 mm/min	ISO 178	MPTS ISO 3167 A	MPa	320
Flexural elongation at max. force	dry, @10 mm/min	ISO 178	MPTS ISO 3167 A	%	2,7
Flexural modulus	dry, @2 mm/min	ISO 178	MPTS ISO 3167 A	GPa	14
Charpy impact strength	dry	ISO 179 1eU	80x10x4mm	kJ/m ²	55
Thermal properties					
Vicat softening temp	VST A	DIN ISO 306	MPTS ISO 3167 A	°C	250
Heat distortion temperature	HDT A	ISO 75	molded sample	°C	245
Continuous service temperature	20.000 h	IEC 60216	MPTS ISO 3167 A	°C	120
Service temperature	during lifetime max. 200h		MPTS ISO 3167 A	°C	160
Electrical properties					
Insulation resistance strip electrode	R25	DIN IEC 60167	MPTS ISO 3167 A	Ω	≤10 ³
Surface resistance	ROB	DIN IEC 60093	Ronde 60x4mm	Ω	<10 ³

Main features

STRUCTURAL

Very strong and stiff parts; low coefficient of thermal expansion. Reduced moment of inertia compared with metal parts. Electrically conductive, suitable for continuous discharging of statically-generated electricity. High dimensionally stable precision parts with low warpage and narrow tolerance range.

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Recommended processing parameters**Predrying**

It is advisable to predry the granules with a suitable dryer immediately before processing. The granule may absorb moisture from the environment.

Dryer type	Temperature °C	Drying time in h
Dehumidifying dryer	75	6 - 16
Vacuum Dryer	105	4 - 6

Processing

Zone 1	°C	290 - 310
Zone 2	°C	290 - 310
Zone 3	°C	290 - 310
Nozzle	°C	280 - 300
Mold	°C	90 - 120
Melt temperature	°C	290

In general this product can be processed on conventional injection moulding machines while observing the usual technical guidelines. Any added fibrous materials or fillers may have an abrasive effect. In this case the cylinder and screw should be protected against wear as is usual in the processing of reinforced thermoplastic materials. Lengthy dwell times for the melts in the cylinder should be avoided. Lower the temperatures during interruptions!

Delivery form & storage

Unless indicated otherwise, the material is delivered as 3mm long pellets in sealed bags on pallets. Preferably storage should be effected in dry and normally temperatured rooms.

Additional information

During processing the moisture level should not exceed 0.05%, otherwise molecular degradation and surface defects (e.g. smearing) may occur. Excessively high predrying temperatures may cause discoloration. The processing notes provided merely represent a recommendation for general use. Due to the large variety of machines, geometries and volumes of parts, etc., it may be necessary to employ different settings according to the specific application. Please contact us for further information.

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